Mechanic Specialization - 6

WEEK 6: INTEGRATION, DIAGNOSTICS & CAPSTONE

Theme: *"Bringing it all together - from diagnosis to complete vehicle service"*

DAY 56: MECHANICAL DIAGNOSTICS METHODOLOGY

MORNING SESSION: THEORY (3 Hours)

Hour 1: Systematic Troubleshooting Approach (9:00 AM - 10:00 AM)

- Why guessing wastes time and money
- The 8-step diagnostic process for mechanical problems
- Customer complaint to root cause: the journey
- Gathering information: what, when, where, how often
- Verifying the problem: reproduce the symptom
- Separating symptoms from root causes
- Using all your senses: look, listen, smell, feel
- Documentation: recording findings at each step

Hour 2: Mechanical Problem Categories (10:00 AM - 11:00 AM)

- Noise diagnosis: location, frequency, conditions
- Vibration diagnosis: speed-related vs load-related
- Performance problems: power loss, hesitation, surging
- Fluid leaks: identifying source and severity
- Wear-related problems: gradual vs sudden failure
- Using process of elimination effectively
- Common failures by vehicle make/model
- When to seek technical service bulletins (TSBs)

Hour 3: Diagnostic Tools & Techniques (11:00 AM - 12:00 PM)

- Visual inspection: 50% of diagnosis is looking
- Test drives: structured approach to reproducing problems
- Mechanical measurements: finding wear limits
- Scan tool data: using mechanical sensor data
- Compression and leakdown tests: engine health
- Pressure tests: cooling, fuel, brake systems

- Sound diagnosis: using mechanic's stethoscope
- Creating diagnostic flowcharts for common problems

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1: Noise Diagnosis Workshop (1:00 PM - 2:00 PM)

- Work on vehicles with actual mechanical problems
- Use stethoscope to locate engine noises
- Identify: rod knock, piston slap, valve train noise
- Diagnose bearing noises in rotating components
- Determine if noise is engine, transmission, or drivetrain
- Use process of elimination (remove belts, etc.)
- Document findings and root cause
- **Pass criteria: correctly diagnose 3 noise problems**

Hour 2: Vibration Diagnosis Case Studies (2:00 PM - 3:00 PM)

- Test drive vehicles with vibration complaints
- Determine vibration speed/condition relationship
- Isolate: engine mount, driveline, tire/wheel balance
- Use dial indicator to check runout
- Check for loose or worn components
- Road test before and after repairs
- Create diagnostic report with recommendations
- **Pass criteria: diagnose vibration root cause**

Hour 3: Performance Problem Diagnosis (3:00 PM - 4:00 PM)

- Diagnose loss of power complaint
- Perform compression test and analyze results
- Check fuel pressure under load
- Inspect air intake for restrictions
- Check exhaust for restrictions
- Test turbo boost pressure (if equipped)
- Use scan tool for mechanical sensor data
- **Pass criteria: complete performance diagnostic procedure**

DAY 57: PREVENTIVE MAINTENANCE EXECUTION

MORNING SESSION: THEORY (3 Hours)

Hour 1: Comprehensive PM Inspection (9:00 AM - 10:00 AM)

- Preventive maintenance vs breakdown repairs
- Service intervals: time vs mileage-based
- Major service vs minor service differences
- Multi-point inspection checklist (40+ points)
- Under hood inspection: all fluids, belts, hoses
- Under vehicle inspection: leaks, wear, damage
- Tire inspection: tread, pressure, wear patterns
- Documentation: customer-facing inspection reports

Hour 2: Scheduled Maintenance Services (10:00 AM - 11:00 AM)

- Engine oil and filter service
- Air filter and cabin filter replacement
- Fuel filter service (diesel especially)
- Coolant flush and replacement
- Transmission fluid service
- Brake fluid flush
- Power steering fluid service
- Differential and transfer case service (4WD/AWD)

Hour 3: Additional PM Services (11:00 AM - 12:00 PM)

- Spark plug replacement intervals
- Timing belt/chain inspection and replacement
- Drive belt inspection and replacement
- Battery testing and replacement
- Wiper blade replacement
- Light bulb inspection and replacement
- Brake inspection and service
- Suspension and steering lubrication

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1: Complete Multi-Point Inspection (1:00 PM - 2:00 PM)

- Perform comprehensive 40-point inspection
- Check all fluid levels and conditions

- Inspect all belts and hoses
- Check battery voltage and load test
- Inspect tires: tread depth, pressure, wear
- Under vehicle inspection on lift
- Test all lights and wipers
- **Pass criteria: complete inspection checklist**

Hour 2: Major Service Execution Part 1 (2:00 PM - 3:00 PM)

- Engine oil and filter change
- Air filter replacement
- Cabin filter replacement
- Top up all fluids
- Check and adjust tire pressures
- Inspect and clean battery terminals
- Lubricate door hinges and locks
- **Pass criteria: complete basic service items**

Hour 3: Major Service Execution Part 2 (3:00 PM - 4:00 PM)

- Coolant flush and refill
- Transmission fluid service
- Brake fluid test and flush (if needed)
- Replace spark plugs
- Inspect and adjust drive belts
- Reset service reminder light
- Complete service sticker and records
- **Pass criteria: major service completion**

DAY 58: CAPSTONE PROJECT - DAY 1

MORNING SESSION: PROJECT BRIEFING & ASSESSMENT (3 Hours)

Hour 1: Capstone Project Introduction (9:00 AM - 10:00 AM)

- Capstone project overview: real-world vehicle service
- Vehicle assignment: each student/group gets a vehicle
- Customer complaint review: understanding the problem
- Initial vehicle walkthrough and documentation
- Taking "before" photos of vehicle condition

- Safety briefing: working independently with supervision
- Tools and parts availability
- Timeline expectations: 2 days to complete

Hour 2: Initial Diagnostic Assessment (10:00 AM - 11:00 AM)

- Perform customer interview (instructor role-plays)
- Document customer complaints and concerns
- Perform complete visual inspection
- Test drive vehicle and note all symptoms
- List all problems found during inspection
- Prioritize repairs: safety, customer complaint, preventive
- Research specifications and procedures
- **Deliverable: Initial diagnostic report**

Hour 3: Diagnostic Testing & Measurement (11:00 AM - 12:00 PM)

- Perform relevant diagnostic tests
- Engine: compression, leakdown (if needed)
- Scan tool: read codes, live data, freeze frame
- Measure wear: brakes, tires, suspension
- Test systems: fuel pressure, cooling, battery
- Document all measurements and findings
- Compare to specifications
- **Deliverable: Complete diagnostic test results**

AFTERNOON SESSION: CAPSTONE WORK (6 Hours - Extended Session)

Hours 1-2: Repair Planning & Parts Ordering (1:00 PM - 3:00 PM)

- Create complete repair list with priorities
- Research parts: part numbers, availability, cost
- Create customer estimate with labor and parts
- Present estimate to instructor (customer role-play)
- Get approval for repairs
- Organize tools needed for all repairs
- Plan work sequence: most efficient order
- **Deliverable: Approved repair estimate**

Hours 3-6: Begin Repair Work (3:00 PM - 7:00 PM)

- Start with highest priority repairs
- Engine work: oil service, filters, spark plugs, etc.
- Cooling system: flush, thermostat, hoses if needed
- Fuel system: filter, cleaning
- Begin disassembly for major repairs if needed
- Document work with photos at each stage
- Keep workspace clean and organized
- **Deliverable: Significant progress on repairs**

DAY 59: CAPSTONE PROJECT - DAY 2

MORNING SESSION: CAPSTONE WORK CONTINUATION (3 Hours)

Hour 1: Complete Engine/Powertrain Repairs (9:00 AM - 10:00 AM)

- Finish any engine-related repairs
- Transmission service if needed
- Timing belt replacement (if applicable)
- Belt and hose replacement
- All fluid services completed
- Torque all fasteners to specification
- Verify all components installed correctly

Hour 2: Chassis & Brake System Repairs (10:00 AM - 11:00 AM)

- Suspension component replacement if needed
- Brake pad/rotor replacement
- Wheel bearing service if required
- Tire rotation and balancing
- Steering system repairs
- All safety-critical systems addressed
- Double-check all work

Hour 3: System Testing & Verification (11:00 AM - 12:00 PM)

- Start engine and check for leaks
- Verify all warning lights off
- Test all systems: brakes, steering, cooling
- Clear diagnostic codes if applicable
- Road test vehicle thoroughly

- Verify customer complaint resolved
- Re-inspect all repair work

AFTERNOON SESSION: COMPLETION & PRESENTATION (3 Hours)

Hour 1: Final Inspection & Quality Check (1:00 PM - 2:00 PM)

- Perform complete post-repair inspection
- Verify all repairs completed correctly
- Check for any leaks or loose components
- Test drive final verification
- Clean vehicle (wash if time permits)
- Take "after" photos
- Complete all documentation
- **Deliverable: Quality inspection checklist**

Hour 2: Documentation & Invoicing (2:00 PM - 3:00 PM)

- Complete detailed service report
- Document all parts used with part numbers
- List all labor operations performed
- Create final invoice for customer
- Update service history records
- Place service reminder sticker
- Prepare vehicle for customer delivery
- **Deliverable: Complete service documentation**

Hour 3: Customer Delivery & Presentation (3:00 PM - 4:00 PM)

- Each student presents their completed work
- Explain customer complaint and diagnosis
- Walk through repairs performed
- Show before/after photos
- Explain invoice and warranty information
- Answer questions from instructor (as customer)
- Instructor evaluation and feedback
- **Deliverable: Professional customer presentation**

DAY 60: FINAL ASSESSMENT & CERTIFICATION

MORNING SESSION: COMPREHENSIVE WRITTEN EXAM (3 Hours)

Hour 1: Theory Examination Part 1 (9:00 AM - 10:00 AM)

- Engine systems: components, operation, diagnosis (20 questions)
- Fuel, cooling, lubrication systems (15 questions)
- Turbocharger and forced induction (10 questions)
- Diesel engine technology (10 questions)
- VVT systems (5 questions)
- **Section Total: 60 questions**

Hour 2: Theory Examination Part 2 (10:00 AM - 11:00 AM)

- Transmission systems: manual, automatic, DSG, CVT (20 questions)
- Clutch systems (10 questions)
- Suspension and steering systems (15 questions)
- Brake systems and ABS (15 questions)
- **Section Total: 60 questions**

Hour 3: Theory Examination Part 3 (11:00 AM - 12:00 PM)

- Emission control systems BS4/BS6 (15 questions)
- Diagnostic procedures and methodology (15 questions)
- Preventive maintenance (10 questions)
- Safety procedures (10 questions)
- Tool usage and measurements (10 questions)
- Customer service scenarios (10 questions)
- **Section Total: 70 questions**
- **GRAND TOTAL: 190 questions (Pass mark: 70%)**

AFTERNOON SESSION: PRACTICAL SKILLS ASSESSMENT (3 Hours)

Hour 1: Practical Exam Station 1 - Measurement & Diagnosis (1:00 PM - 2:00 PM)

- Task 1: Measure cylinder bore with dial bore gauge (10 points)
- Task 2: Measure piston clearance and calculate (10 points)
- Task 3: Use torque wrench correctly on test fixture (10 points)

- Task 4: Diagnose engine noise using stethoscope (10 points)
- Task 5: Interpret compression test results (10 points)
- **Station Total: 50 points**

Hour 2: Practical Exam Station 2 - Service Procedures (2:00 PM - 3:00 PM)

- Task 1: Perform brake pad replacement (15 points)
- Task 2: Replace and bleed hydraulic clutch (15 points)
- Task 3: Perform wheel alignment toe adjustment (15 points)
- Task 4: Service automatic transmission fluid (15 points)
- Task 5: Replace oxygen sensor correctly (10 points)
- **Station Total: 70 points**

Hour 3: Practical Exam Station 3 - Diagnosis & Scan Tool (3:00 PM - 4:00 PM)

- Task 1: Use scan tool to diagnose check engine light (15 points)
- Task 2: Test fuel pressure and interpret results (10 points)
- Task 3: Diagnose ABS warning light with scan tool (15 points)
- Task 4: Test VVT solenoid operation (10 points)
- Task 5: Create professional diagnostic report (10 points)
- **Station Total: 60 points**

PRACTICAL TOTAL: 180 points (Pass mark: 70% = 126 points)

CERTIFICATION CEREMONY (After 4:00 PM)

Final Results & Certification

- Announcement of results
- Individual feedback sessions
- Certificate presentation ceremony
- Group photo with certificates
- Instructor final remarks and advice
- Industry representative speech (if available)
- Career counseling available
- Next steps: job placement assistance

WEEK 6 & OVERALL TRACK OUTCOMES

By the end of Week 6 and Track A completion, students will be able to:

✅ **Apply systematic diagnostic methodology** to mechanical problems
✅ **Diagnose complex mechanical issues** using multiple tools and techniques
✅ **Execute complete preventive maintenance services** professionally
✅ **Perform major vehicle repairs** independently with supervision
✅ **Create professional estimates and documentation**
✅ **Communicate effectively with customers** about repairs
✅ **Work safely and efficiently** in a workshop environment
✅ **Integrate knowledge from all systems** to solve real problems
✅ **Present completed work professionally** with confidence
✅ **Meet industry standards** for NSQF Level 5 mechanical technicians

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TRACK A: MECHANICAL MAINTENANCE - COMPLETE SUMMARY

Total Duration: 30 days (6 weeks)
Total Hours: 180 hours (3+3 daily)

Week-by-Week Recap:

- **Week 1:** Precision tools, measurements, engine disassembly/inspection
- **Week 2:** VVT, diesel engines, turbocharger systems
- **Week 3:** Fuel, cooling, lubrication, emission systems
- **Week 4:** Manual, automatic, CVT, DSG transmissions
- **Week 5:** Suspension, alignment, steering, brakes, ABS
- **Week 6:** Diagnostics, preventive maintenance, capstone project

Career Readiness:

✅ **Ready for positions:** Mechanical Technician, Engine Specialist, Transmission Specialist
✅ **Industry certification:** ASC/Q6808 Automotive Service Technician - Mechanical
✅ **Salary expectation:** ₹18,000 - ₹25,000/month (entry level)
✅ **Growth path:** Senior Technician → Master Technician → Shop Foreman

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TRACK A MECHANICAL MAINTENANCE COMPLETE! 🎓

Would you like me to:

1. Create detailed assessment rubrics for the practical exams?
2. Develop the capstone project evaluation criteria?
3. Create student handbooks for each week?

4. Proceed to develop Track B (Electrical Maintenance) or Track C (Unified Diagnostics)?